

6 Results

This section reports the held-out test results for five-class grading (Section 6.1) and referable-DR screening (Section 6.2), followed by the ablation study, the effect of inference-time choices, training behaviour, visual explanations and computational cost. Unless stated otherwise, all numbers refer to the **[ntest]**-image IDRiD test set and to the five-fold ensemble with test-time augmentation.

6.1 Five-class grading

Table 4 reports the held-out test performance. Cross-validation estimates on the training portion are given for comparison: OOF accuracy **[oofacc]**% and QWK **[oofqwk]** (per-fold accuracy **[foldacc]**%, per-fold QWK **[foldqwk]**).

Table 4: Five-class DR grading on the held-out IDRiD test set ($n = \text{[ntest]}$). Ensemble of five fold models with four-view TTA, decision rule chosen on OOF data. Brackets: 95% bootstrap CI.

Metric	Value	Metric	Value
Accuracy (%)	[acc5] [acc5ci]	Macro precision (%)	[precmacro5]
Balanced accuracy (%)	[bacc5]	Macro recall (%)	[recmacro5]
Quadratic weighted kappa	[qwk5] [qwk5ci]	Macro F1 (%)	[f1macro5] [f1macro5ci]
Cohen’s kappa	[kappa5]	Weighted F1 (%)	[f1w5]
MCC	[mcc5]	Within-one-grade accuracy (%)	[within1]
Macro ROC AUC (OvR)	[auc5]	Macro average precision	[ap5]

Table 5: Per-grade precision, recall and F1 (%) on the test set.

	0 No DR	1 Mild	2 Moderate	3 Severe	4 PDR
Support	[nte0]	[nte1]	[nte2]	[nte3]	[nte4]
Precision	[p0]	[p1]	[p2]	[p3]	[p4]
Recall	[r0]	[r1]	[r2]	[r3]	[r4]
F1	[f0]	[f1]	[f2]	[f3]	[f4]

Table 5 and Fig. 3(a) show where errors occur. *[AUTHOR: Describe the confusion pattern: e.g. whether most errors are between adjacent grades (consistent with within-one-grade accuracy of **[within1]**%), and which grade is weakest. Grade 1 is typically hardest because it is defined by microaneurysms only.]*

6.2 Referable DR screening

Table 6 reports the screening task. The operating threshold was fixed on OOF data before the test set was evaluated; the OOF AUC was **[oofrefauc]**. Figure 3 (right) shows the ROC curves.

Table 6: Referable DR (grade ≥ 2 vs. < 2) on the held-out test set. Brackets: 95% bootstrap CI.

Metric	Value	Metric	Value
ROC AUC	[refauc] [refauc]	PPV (%)	[refppv]
Sensitivity (%)	[refsens] [refsensci]	NPV (%)	[refnpv]
Specificity (%)	[refspec] [refspec]	F1 (%)	[reff1]
Accuracy (%)	[refacc] [refacc]	TP / FN / FP / TN	[tp] / [fn] / [fp] / [tn]

Figure file `figures/confusion.pdf` not found.
Confusion matrices (5-class and referable).

Figure file `figures/roc.pdf` not found.
ROC curves.

Figure 3: Held-out test set. Left: confusion matrices for (a) five-class grading and (b) referable DR (grade ≥ 2). Right: one-vs-rest ROC curves for each grade and for referable DR (bold).

6.3 Ablation study

Table 7 compares the full model with the configurations of Section 5. All runs share the same test images, so the McNemar p -value tests whether per-image correctness differs from the full model.

Table 7: Ablation study on the held-out IDRiD test set. Five-class accuracy, QWK and macro F1; referable-DR AUC, sensitivity and specificity. p : exact McNemar test vs. the full model on five-class correctness.

Configuration	Params (M)	Acc (%)	QWK	F1 (%)	Ref. AUC	Sens. (%)	Spec. (%)
NITSCI (full)	[abfullparams]	[abfullacc]	[abfullqwk]	[abfullf1]	[abfullrefauc]	[abfullrefsens]	[abfullrefspec]
Plain (GAP head)	[abplainparams]	[abplainacc]	[abplainqwk]	[abplainf1]	[abplainrefauc]	[abplainrefsens]	[abplainrefspec]
w/o APTOS pretraining	[abnoaptosparams]	[abnoaptosacc]	[abnoaptosqwk]	[abnoaptosf1]	[abnoaptosrefauc]	[abnoaptosrefsens]	[abnoaptosrefspec]
CLAHE pre-processing	[abclaheparams]	[abclaheacc]	[abclaheqwk]	[abclahef1]	[abclaherefauc]	[abclaherefsens]	[abclaherefspec]
w/o ordinal head	[abnoregparams]	[abnoregacc]	[abnoregqwk]	[abnoregf1]	[abnoregrefauc]	[abnoregrefsens]	[abnoregrefspec]
w/o lesion map	[abnolesionparams]	[abnolesionacc]	[abnolesionqwk]	[abnolesionf1]	[abnolesionrefauc]	[abnolesionrefsens]	[abnolesionrefspec]
w/o cross-attention	[abnocrossparams]	[abnocrossacc]	[abnocrossqwk]	[abnocrossf1]	[abnocrossrefauc]	[abnocrossrefsens]	[abnocrossrefspec]
w/o adaptive gate	[abnogateparams]	[abnogateacc]	[abnogateqwk]	[abnogatef1]	[abnogaterefauc]	[abnogaterefsens]	[abnogaterefspec]

McNemar p vs. full: Plain **[abplainp]**; w/o APTOS **[abnoaptosp]**; CLAHE **[abclahep]**; w/o ordinal **[abnoregp]**; w/o lesion map **[abnolesionp]**; w/o cross-attention **[abnocrossp]**; w/o gate **[abnogatep]**. Parameter counts of “w/o cross-attention” and “w/o gate” include the unused modules, which are kept so that the remaining weights are initialised identically.

[AUTHOR: Interpret Table 7 honestly: name the components whose removal lowers accuracy/QWK by more than the width of the bootstrap interval or with McNemar $p < 0.05$, and state plainly which components show no measurable effect at this test-set size. Do not claim a component “improves” performance if its p -value is large.]

6.4 Inference-time choices

Table 8 isolates the effect of the decision rule, TTA and fold ensembling without retraining. The decision rule was selected on OOF accuracy; test values are shown only to document the outcome of that choice.

6.5 Training behaviour

Supplementary Fig. S2 shows the training loss and validation metrics for each fold. *[AUTHOR: Comment on convergence, the epoch at which early stopping triggered, and the spread between folds.]*

Table 8: Effect of inference-time choices. Top: decision rules (OOF accuracy was used for selection). Bottom: single fold models vs. ensemble, with and without TTA (mean $\pm$ SD over five folds for single models).

Decision rule	OOF acc. (%)	Test acc. (%)	Test QWK
Argmax of p	[oofargmax]	[testargmax]	[testargmaxqwk]
Rounded score s	[oofrounded]	[testrounded]	[testroundedqwk]
Optimised thresholds on s	[oofopt]	[testopt]	[testoptqwk]

Prediction	Test acc. (%)	Test QWK
Single fold, TTA	[singleacc]	[singleqwk]
5-fold ensemble, no TTA	[notaacc]	[notaqwk]
5-fold ensemble + TTA	[ttaacc]	[ttaqwk]

6.6 Visual explanations

Figure 4 shows Grad-CAM maps for the predicted grade and the learned lesion-attention map m for test images, together with the gate value g (Eq. 5). Because m is learned without lesion annotations, these maps are qualitative only; we do not claim that they localise lesions. *[AUTHOR: Describe what the maps show, e.g. whether high attention falls on haemorrhages or exudates, and whether g differs between low and high grades. A quantitative check against the IDRiD lesion masks (e.g. pointing-game accuracy) would strengthen this section.]*

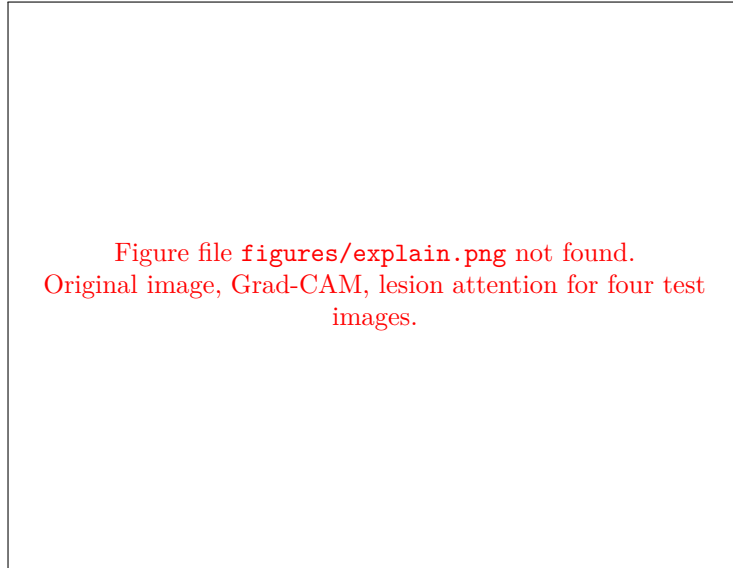


Figure 4: Test images (top), Grad-CAM for the predicted grade (middle), and the learned lesion-attention map with gate value g (bottom), from the fold-1 model.

6.7 Computational cost

The full model has 21.25 M parameters and requires 23.0 GFLOPs per 448×448 image (Table 2). Measured inference time on the [gpu] was [mspi] ms per image per model and view, so the 20-prediction ensemble costs about twenty times this value per image. *[AUTHOR: State whether this is acceptable for screening throughput, and note that a single fold model without TTA is the cheaper option whose accuracy is given in Table 8.]*